

Summer 2025

ENERGY CONVERSION-I

EEE 221

LECTURE-05



Inspiring Excellence

Transformers

Chapter 2 Transformers

- Types and construction of transformers
- The ideal transformer
- Theory of operation of real single-phase transformers
- Equivalent circuit of a transformer
- Transformer voltage regulation and efficiency
- Transformer taps and voltage regulation
- The autotransformer
- **Three-phase transformer**
- **Instrument transformers**



High Voltage Side
Delta Connected

Low Voltage Side
Y-Connected

Three-phase transformer – three single-phase transformer banks

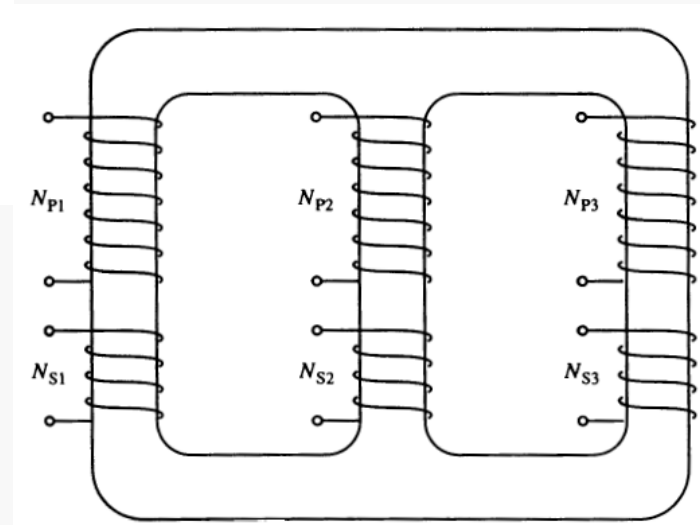
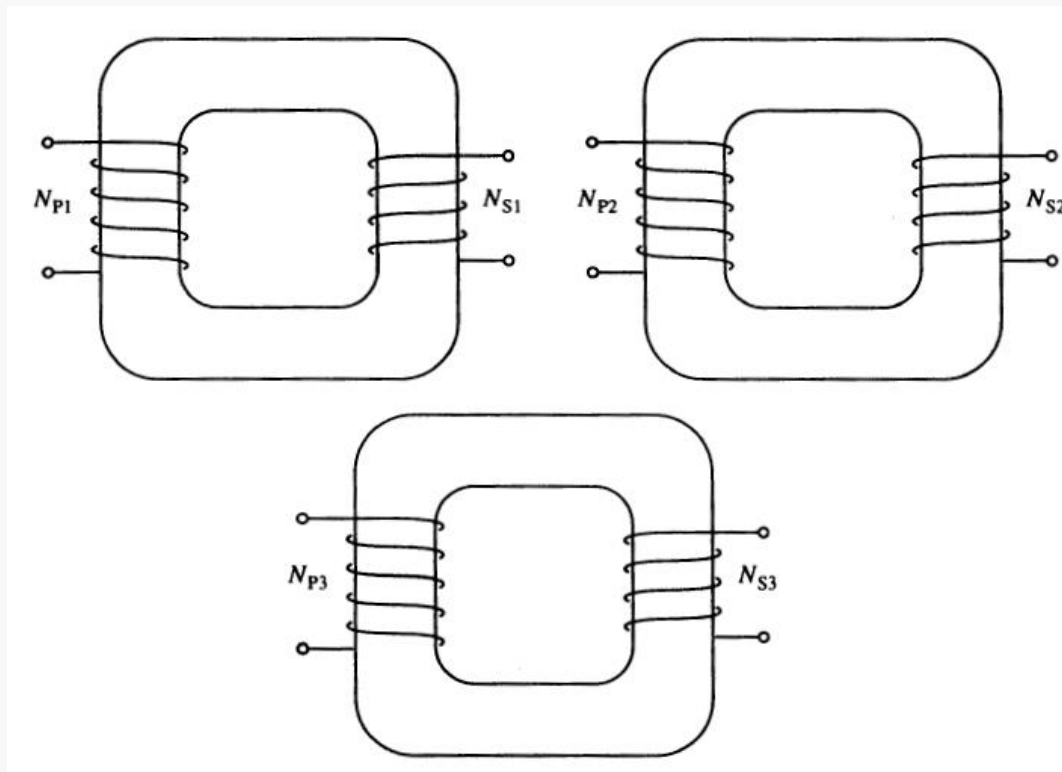


Figure 2-35

Three types of terminal connection

A three-phase transformer consists of three transformers, either separate or combined on one core. The primaries and secondaries of any three-phase transformer can be independently connected in either a wye (Y) or a delta (Δ). This gives a total of four possible connections for a three-phase transformer bank:

1. Wye–wye (Y–Y)
2. Wye–delta (Y– Δ)
3. Delta–wye (Δ –Y)
4. Delta–delta (Δ – Δ)

Wye-wye connection

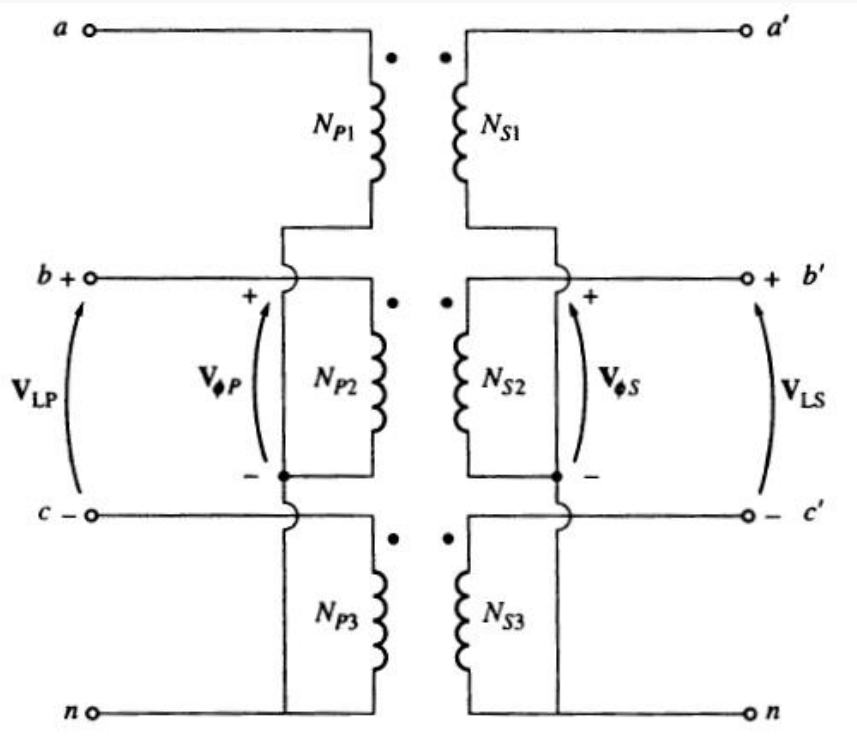
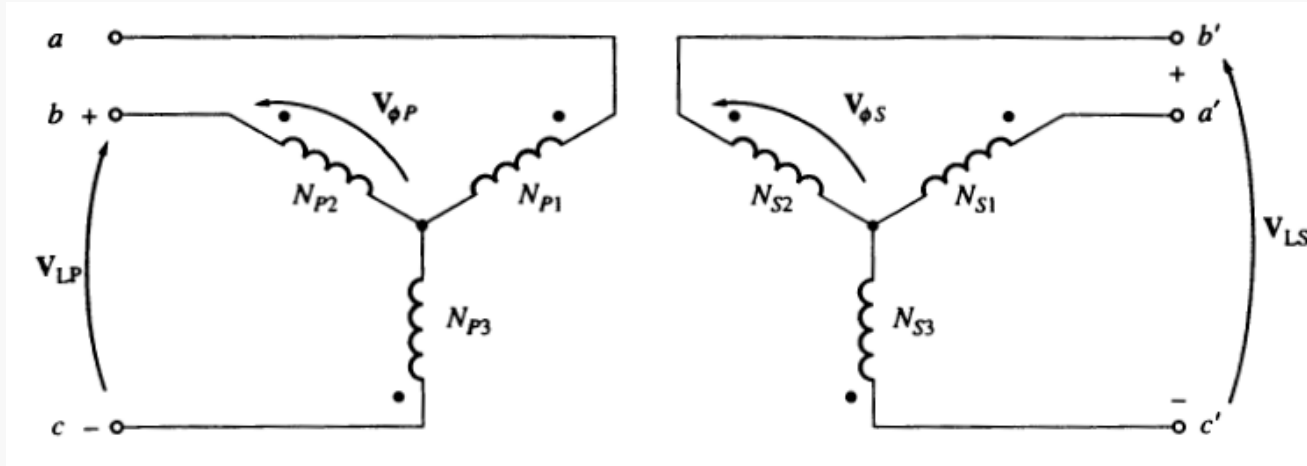


Figure 2-37 (a)

Wye-delta connection

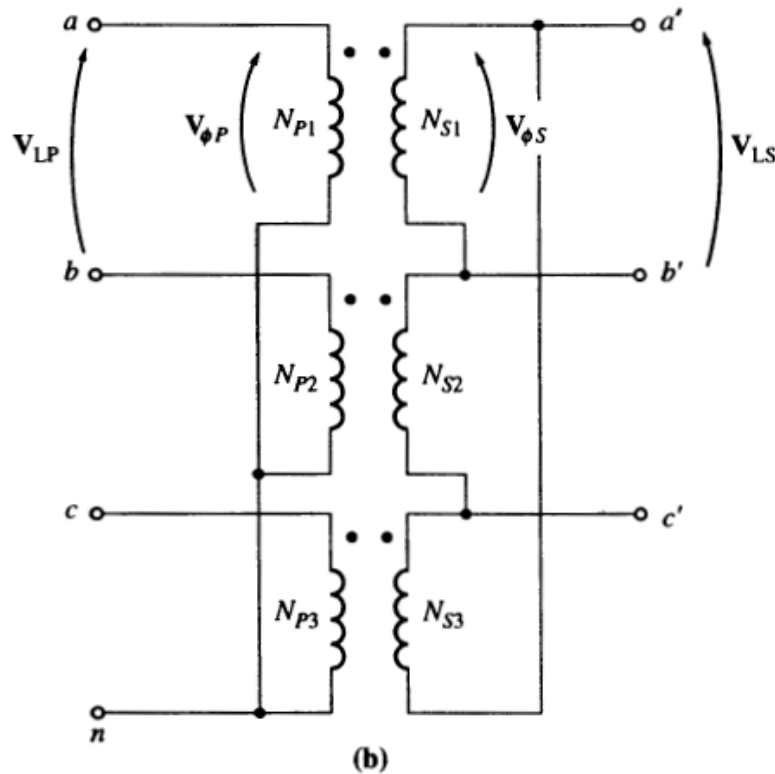
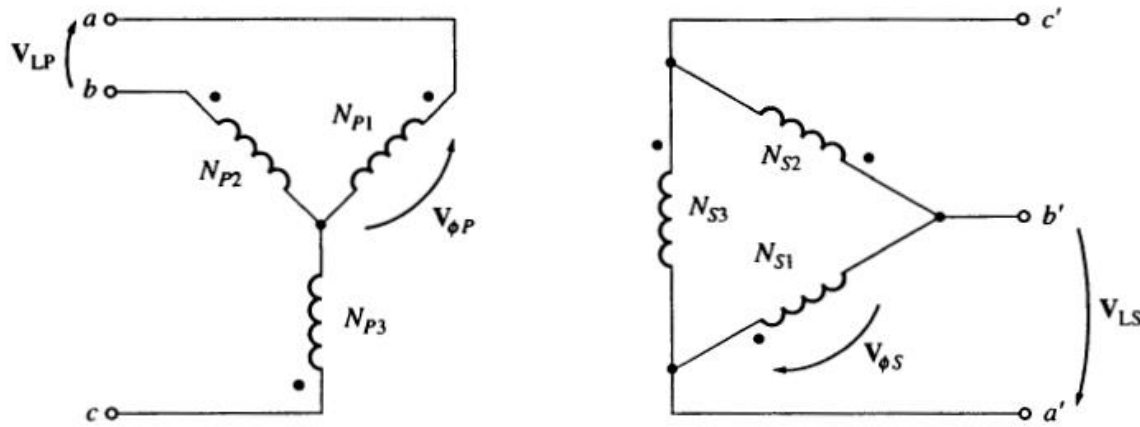
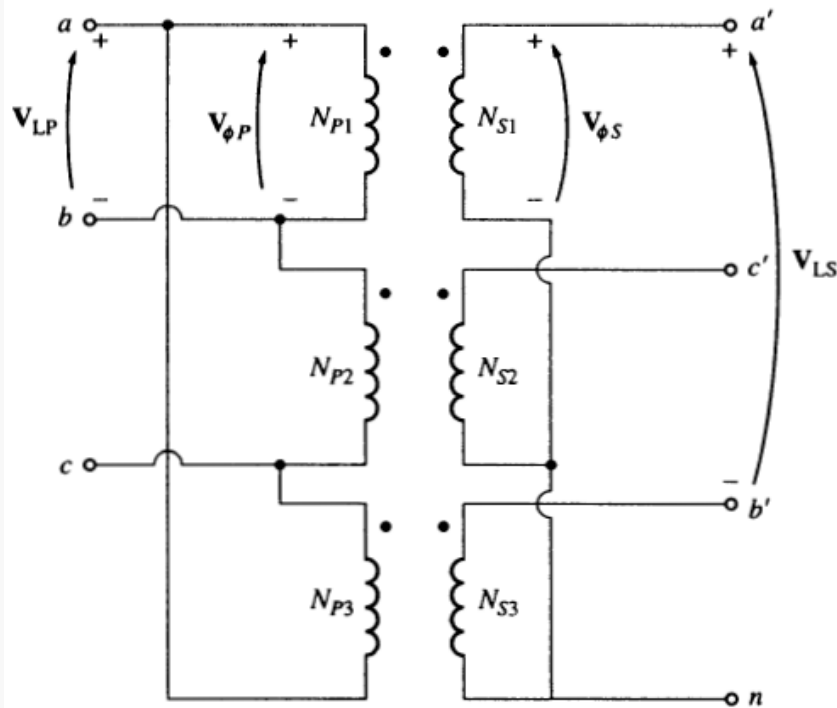
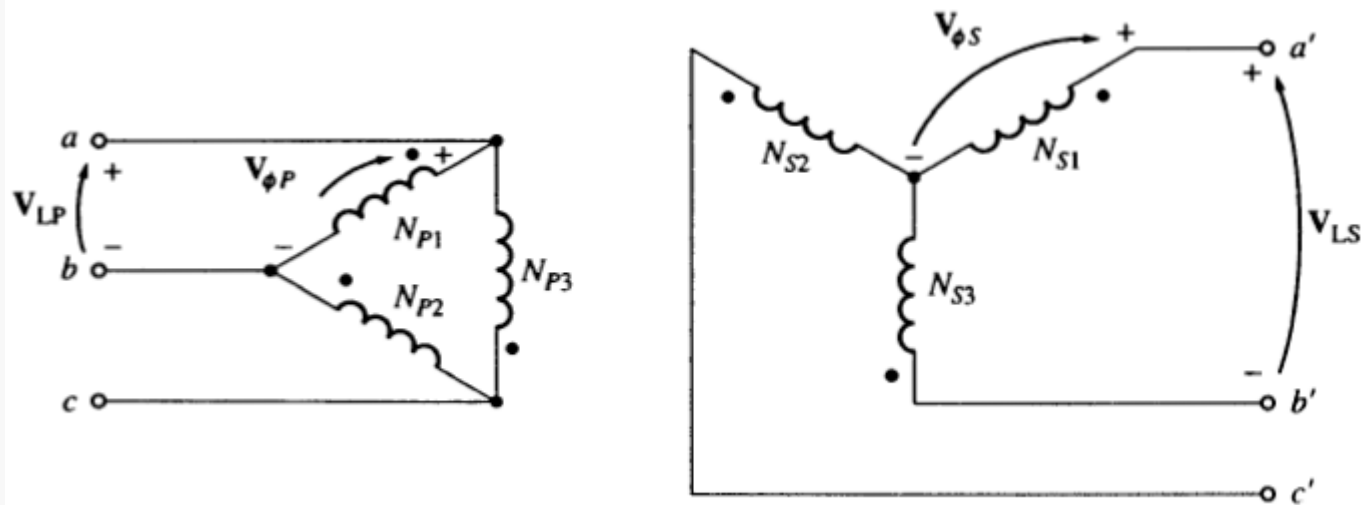


Figure 2-37 (b)

Wye-delta connection summarize

1. No 3rd harmonic component problem:
2. There are phase difference between each sides
3. The secondary voltage will lag the primary voltage 30 degrees

Delta-wye connection



$$\frac{V_{LP}}{V_{LS}} = \frac{V_{\phi P}}{\sqrt{3}V_{\phi S}} = \frac{a}{\sqrt{3}}$$

Figure 2-37 (c)

Delta-wye connection

- Common connection:
 - *Used on three-wire (delta) to four wire (wye)*
 - *Used to isolate ground on wye side from source ground on delta side*

Delta-wye or wye-delta?

- Common for wye-delta step-up transformer banks in generating plants
- Common for delta-wye step-down banks in substation

Delta-delta connection

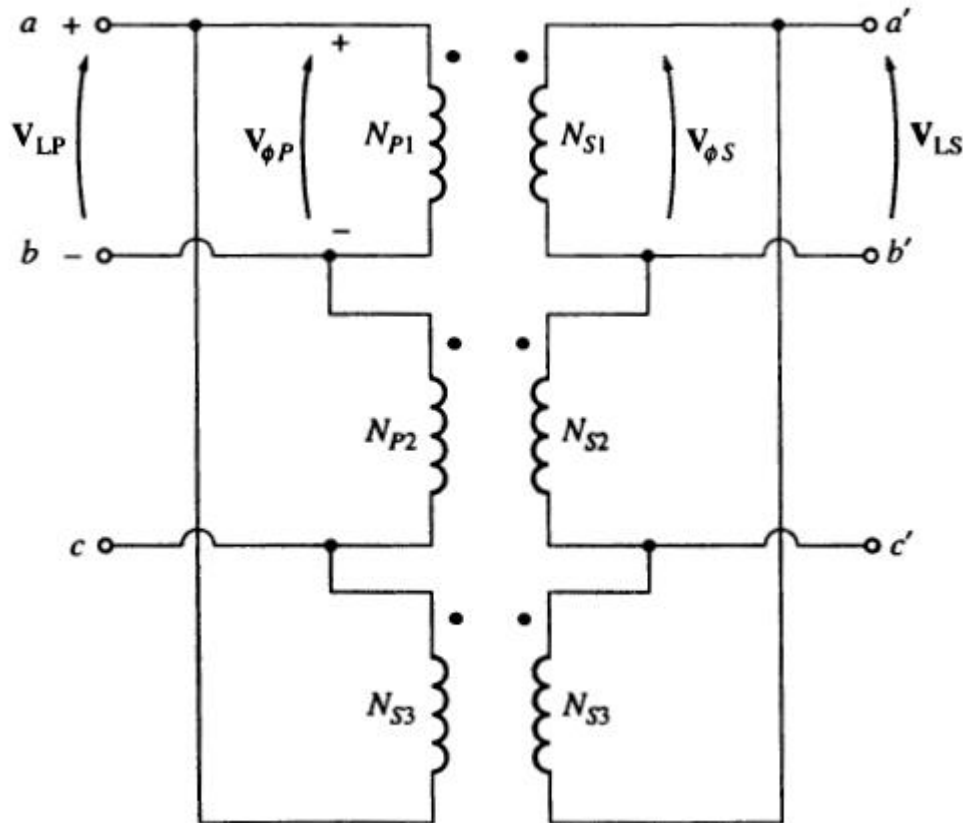
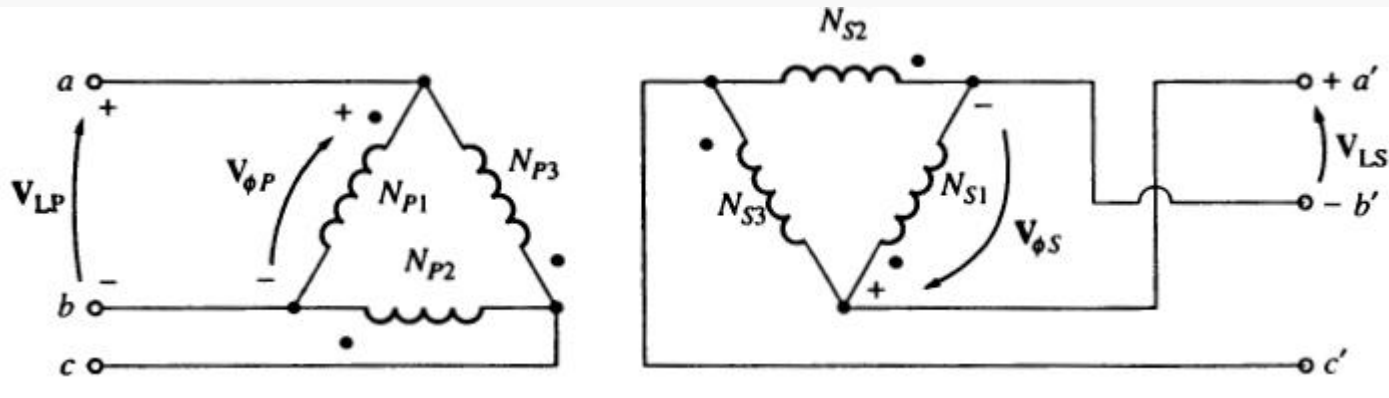


Figure 2-37 (d)

Instrument transformer

- Potential transformer (PT)
- Current transformer (CT)



PT

CT

Instrument transformer

A **potential transformer** has a high-voltage primary, low-voltage secondary, and very low power rating. It is used to provide an accurate voltage samples to instruments monitoring the power system. **The rated secondary voltage usually 110 Volt.**

A **current transformer** samples the current in a line and reduces it to a safe and measurable level. Such transformer consists of a secondary winding wrapped around a ferromagnetic ring with a single primary line (that may carry a large current)running through its center. The ring holds a small sample of the flux from the primary line. That flux induces a secondary voltage.

Current Transformer

Windings in current transformers are loosely coupled: the mutual flux is much smaller than the leakage flux. The voltage and current ratios do not apply although the secondary current is directly proportional to the primary.

Current transformers **must be short-circuited** at all times since very high voltages can appear across their terminals.

Current transformer ratings are given as ratios of primary to secondary current.

A typical ratio might be 600:5, 800:5 or 1000:5. A 5-A rating is standard on the secondary of a current transformer

